

# Project: MicroLend

## *Microfinance & NBFC Loan Management System*

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Domain: Banking & Financial Services

### 1. Introduction

MicroLend is a web-based microfinance and non-banking financial company loan management system designed for microfinance institutions, NBFCs, and rural lending cooperatives to manage borrower onboarding, group lending models, loan origination, disbursements, repayment collections, field officer activities, and portfolio quality monitoring. The system enables borrowers to apply for loans and track repayment schedules, field officers to conduct centre meetings, record collections, and manage group formation, credit officers to assess applications and approve disbursements, and administrators to configure loan products, interest rate models, and collection rules.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

### Actors / Users

- **Borrower:** Applies for a loan, views repayment schedule, and tracks loan account status
- **Field Officer:** Forms borrower groups, conducts centre meetings, and records repayment collections
- **Credit Officer:** Assesses loan applications, performs credit checks, and approves disbursements
- **Branch Manager:** Oversees branch portfolio quality, delinquency cases, and field officer productivity
- **Collections Officer:** Manages overdue accounts, recovery actions, and restructuring workflows
- **NBFC Admin:** Configures loan products, interest models, collection rules, and user roles

### 2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.

- **2.2 Borrower Onboarding & KYC:** Manages borrower profiles, KYC document verification, and credit bureau checks.
- **2.3 Group & Centre Management:** Manages borrower groups, joint liability group formation, and centre meeting schedules.
- **2.4 Loan Origination & Approval:** Handles loan applications, credit scoring, approval workflows, and sanction letters.
- **2.5 Loan Disbursement & Account Management:** Manages loan disbursals, account creation, and repayment schedule generation.
- **2.6 Repayment Collection & Delinquency Management:** Records collections, manages overdue accounts, and tracks recovery actions.
- **2.7 Portfolio Analytics & Reporting:** Provides dashboards for PAR, disbursement volumes, collection efficiency, and branch performance.
- **2.8 Notifications & Alerts:** Repayment reminders, overdue escalations, meeting schedules, and disbursement confirmations.

### 3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

### 4. Module-Wise Design

#### 4.1 Identity & Access Management Module

##### Features

- Borrower and staff registration, login, and role-based access control
- Branch-scoped session management and full audit logging

##### Entities

- **User**
  - UserID
  - Name
  - Role (Borrower/FieldOfficer/CreditOfficer/BranchManager/Collections/Admin)
  - Email

- Phone
- BranchID
- Status (Active/Inactive/Suspended)
- **AuditLog**
  - AuditID
  - UserID
  - Action
  - Module
  - Timestamp

## 4.2 Borrower Onboarding & KYC Module

### Features

- Register borrowers with demographic, income, and identity details
- Capture KYC documents and record internal credit assessment scores

### Entities

- **Borrower**
  - BorrowerID
  - Name
  - DateOfBirth
  - Gender
  - NationalIDNumber
  - Village
  - District
  - Phone
  - Occupation
  - MonthlyIncome
  - BankAccountNumber
  - Status (Active/Blacklisted/Deceased)

- **BorrowerKYC**
  - KYCID
  - BorrowerID
  - DocumentType (NationalID/VoterID/Passport/UtilityBill)
  - DocumentRef
  - VerifiedByID
  - VerificationDate
  - Status (Pending/Verified/Rejected/Expired)
- **CreditAssessment**
  - AssessmentID
  - BorrowerID
  - AssessedByID
  - AssessmentDate
  - InternalCreditScore
  - DebtBurdenRatio
  - Recommendation (Eligible/NotEligible/Conditional)
  - Remarks

## 4.3 Group & Centre Management Module

### Features

- Form joint liability borrower groups and assign to field centres
- Schedule weekly or monthly centre meetings and track attendance

### Entities

- **BorrowerGroup**
  - GroupID
  - GroupName
  - CentreID
  - FieldOfficerID

- FormationDate
- MemberCount
- JointLiabilityEnabled
- Status (Active/Dissolved/Suspended)
- **Centre**
  - CentreID
  - CentreName
  - BranchID
  - FieldOfficerID
  - Village
  - MeetingDay (Monday/Tuesday/Wednesday/Thursday/Friday/Saturday)
  - MeetingTime
  - Status (Active/Inactive)
- **CentreMeeting**
  - MeetingID
  - CentreID
  - MeetingDate
  - ConductedByID
  - AttendanceCount
  - CollectionAmount
  - Status (Scheduled/Conducted/Missed/Postponed)

## 4.4 Loan Origination & Approval Module

### Features

- Process individual and group loan applications with configurable product parameters
- Enforce credit policy rules, multi-level approval workflows, and sanction letter generation

### Entities

- **LoanApplication**

- ApplicationID
- BorrowerID
- GroupID
- LoanProductID
- RequestedAmount
- Purpose
- ApplicationDate
- CreditOfficerID
- Status (Draft/Submitted/UnderReview/Approved/Rejected/Disbursed)
- **LoanProduct**
  - ProductID
  - ProductName
  - Category (Individual/GroupLending/MSME/Agricultural)
  - MinAmount
  - MaxAmount
  - TenureMonths
  - InterestRatePercent
  - InterestType (Flat/Reducing)
  - ProcessingFeePercent
  - Status (Active/Discontinued)
- **SanctionLetter**
  - SanctionID
  - ApplicationID
  - SanctionedAmount
  - InterestRate
  - Tenure
  - EMIAmount
  - DisbursalConditions

- IssuedDate
- AcceptedByBorrower
- Status (Issued/Accepted/Lapsed)

## 4.5 Loan Disbursement & Account Management Module

### Features

- Create loan accounts post-sanction and generate repayment schedules
- Track disbursement tranches and outstanding principal balance

### Entities

- **LoanAccount**
  - LoanAccountID
  - ApplicationID
  - BorrowerID
  - ProductID
  - DisbursedAmount
  - DisbursementDate
  - TotalInterest
  - TotalRepayable
  - OutstandingPrincipal
  - Status (Active/Closed/NPA/WrittenOff)
- **RepaymentSchedule**
  - ScheduleID
  - LoanAccountID
  - InstallmentNumber
  - DueDate
  - PrincipalDue
  - InterestDue
  - TotalDue

- Status (Pending/Paid/Overdue/Waived)

## 4.6 Repayment Collection & Delinquency Management Module

### Features

- Record cash and bank repayment collections against due instalments
- Track overdue accounts, manage restructuring workflows, and log recovery actions

### Entities

- **CollectionRecord**
  - CollectionID
  - LoanAccountID
  - ScheduleID
  - CollectedAmount
  - CollectionDate
  - CollectedByID
  - Mode (Cash/BankTransfer/CentreCollection)
  - Status (Received/Partial/Excess)
- **DelinquencyCase**
  - CaseID
  - LoanAccountID
  - DPD (DaysPastDue)
  - PARBucket (PAR30/PAR60/PAR90/PAR180)
  - AssignedCollectionsOfficerID
  - OpenedDate
  - Action (FieldVisit/LegalNotice/Restructuring/WriteOff)
  - Status (Open/InProgress/Resolved/WrittenOff)

## 4.7 Portfolio Analytics & Reporting Module

### Features



- PAR ratio, disbursement volume, and collection efficiency dashboards by branch and product
- Field officer productivity, overdue bucket migration, and loan cycle analysis reports

## Entities

- **PortfolioReport**
  - ReportID
  - Scope (Branch/Product/FieldOfficer/Period)
  - Metrics (ActiveLoanCount, TotalOutstanding, DisbursementValue, CollectionRate, PAR30, PAR90, NPAPercent)
  - GeneratedDate

## 4.8 Notifications & Alerts Module

### Features

- Repayment due reminders and overdue escalation alerts for borrowers and field officers
- Centre meeting schedule notifications, disbursement confirmations, and NPA migration warnings for branch managers

### Entities

- **Notification**
  - NotificationID
  - UserID
  - Message
  - Category (Repayment/Disbursement/Meeting/Delinquency/Compliance)
  - Status (Unread/Read/Dismissed)
  - CreatedDate

## 5. Deployment Strategy

- Local: Angular/React frontend, Spring Boot/.NET Core backend, local DB
- Production: Cloud or on-premises with secure APIs and stateless services

## 6. Database Design

### Tables:

- **User**(UserID, Name, Role, Email, Phone, BranchID, Status)
- **AuditLog**(AuditID, UserID, Action, Module, Timestamp)
- **Borrower**(BorrowerID, Name, DateOfBirth, Gender, NationalIDNumber, Village, District, Phone, Occupation, MonthlyIncome, BankAccountNumber, Status)
- **BorrowerKYC**(KYCID, BorrowerID, DocumentType, DocumentRef, VerifiedByID, VerificationDate, Status)
- **CreditAssessment**(AssessmentID, BorrowerID, AssessedByID, AssessmentDate, InternalCreditScore, DebtBurdenRatio, Recommendation, Remarks)
- **BorrowerGroup**(GroupID, GroupName, CentreID, FieldOfficerID, FormationDate, MemberCount, JointLiabilityEnabled, Status)
- **Centre**(CentreID, CentreName, BranchID, FieldOfficerID, Village, MeetingDay, MeetingTime, Status)
- **CentreMeeting**(MeetingID, CentreID, MeetingDate, ConductedByID, AttendanceCount, CollectionAmount, Status)
- **LoanProduct**(ProductID, ProductName, Category, MinAmount, MaxAmount, TenureMonths, InterestRatePercent, InterestType, ProcessingFeePercent, Status)
- **LoanApplication**(ApplicationID, BorrowerID, GroupID, LoanProductID, RequestedAmount, Purpose, ApplicationDate, CreditOfficerID, Status)
- **SanctionLetter**(SanctionID, ApplicationID, SanctionedAmount, InterestRate, Tenure, EMIAmount, DisbursalConditions, IssuedDate, AcceptedByBorrower, Status)
- **LoanAccount**(LoanAccountID, ApplicationID, BorrowerID, ProductID, DisbursedAmount, DisbursementDate, TotalInterest, TotalRepayable, OutstandingPrincipal, Status)
- **RepaymentSchedule**(ScheduleID, LoanAccountID, InstallmentNumber, DueDate, PrincipalDue, InterestDue, TotalDue, Status)
- **CollectionRecord**(CollectionID, LoanAccountID, ScheduleID, CollectedAmount, CollectionDate, CollectedByID, Mode, Status)
- **DelinquencyCase**(CaseID, LoanAccountID, DPD, PARBucket, AssignedCollectionsOfficerID, OpenedDate, Action, Status)
- **PortfolioReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

## 7. User Interface Design

- **Borrower Portal:** Loan account summary, repayment schedule, collection receipt history, application tracker
- **Field Officer App:** Assigned centre list, meeting log form, collection entry screen, group roster
- **Credit Officer Workbench:** Application queue, credit assessment form, sanction letter generator, approval dashboard
- **Branch Manager Console:** Portfolio quality dashboard, PAR bucket breakdown, field officer productivity, delinquency case tracker
- **Collections Dashboard:** Overdue account list, DPD aging report, recovery action log, restructuring queue
- **Admin Console:** Loan product builder, interest rate configuration, collection rule setup, branch and centre management

## 8. Non-Functional Requirements

- **Performance:** Support 20,000+ concurrent users across field officers, borrowers, and branch staff
- **Security:** Borrower PII protection, bank account data encryption, RBAC, and full audit trails for all financial entries
- **Scalability:** Multi-branch and multi-state deployment with branch-scoped portfolio data partitioning
- **Availability:** 99.9% uptime with collection entry and loan account modules prioritised for high availability
- **Maintainability:** Configurable loan products, interest calculation rules, and delinquency escalation workflows
- **Observability:** Real-time PAR dashboards, collection efficiency metrics, and delinquency bucket migration alerts

## 9. Assumptions & Constraints

- Phase 1 excludes integration with external credit bureau APIs and government ID verification services
- Disbursements are recorded as internal ledger entries; bank API-based fund transfer is deferred to Phase 2
- All notifications are in-app only; SMS repayment reminder integrations are out of Phase 1 scope

- Credit scoring uses internally defined rule-based parameters; ML-based scoring is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling